

AHB

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(Advanced High Performance Bus)

- ✓ Burst transfers → ^{sending} multiple data blocks at same time
- ✓ Split transfer
- ✓ Single clock edge operation
- ✓ wider Data Bus configuration (64/128 bit)

* AHB Transfers

→ Types of Transfer - HTRANS[1:0]
signal 2 bit

00 → idle (no new transfer)

01 → Busy (allow master to insert IDLE cycle in middle of Burst Transfer)

10 → NONSEQ (indicate a new transfer)

11 → SEQ (indicate a remaining transfer in a burst)

→ Transfer size - $HSIZE[2:0]$

[2] [1] [0]

0 0 0

→ 8 bit (Byte)

0 0 1

→ 16 bit halfword

0 1 0

→ 32 bit word

0 1 1

→ 64

1 0 0

→ 128

4 word the

1 0 1

→ 256

1 1 0

→ 512

1 1 1

→ 1024

1KB

* Burst Transfer

Burst size indicates = no. of beats in the burst

incrementing Bursts

↳ can be of any length but address must not cross 1KB

→ increment previous address

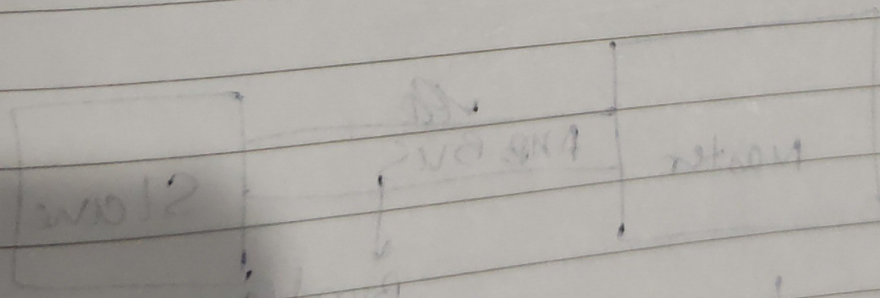
wrapping bursts

Burst → Combines multiple beats

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Lock 4 → means transferring 4 beats in 1 Burst

like to send 32 bit data in 4 Bytes
∴ 8 parts split



only Master can control the bus at a time to send data

provided the address control signal data bus

AHB 3 main components

① Manager (Master)

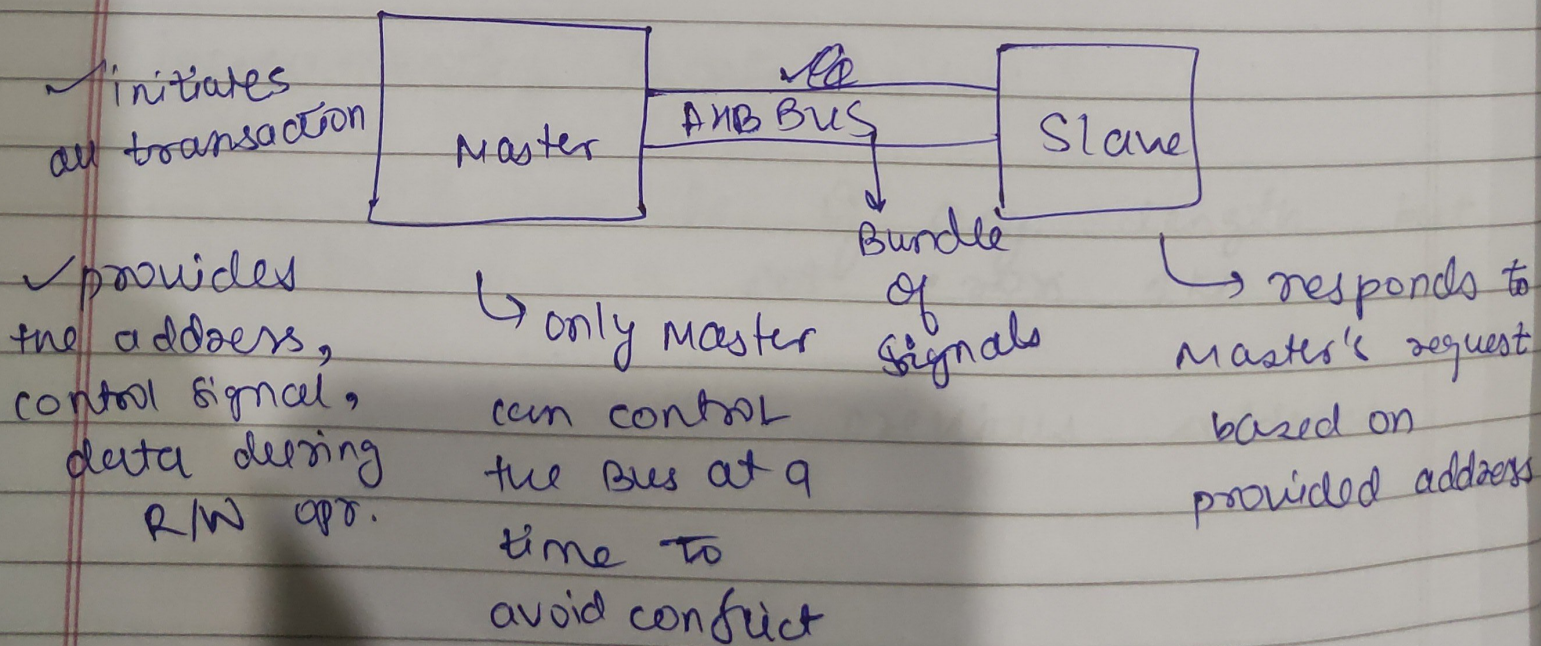
initiate transactions by issuing read and write commands

② Slave

responds to Manager requests and provide or store data.

③ Interconnects

↳ Handle data routing & arbitration between Master & Slave

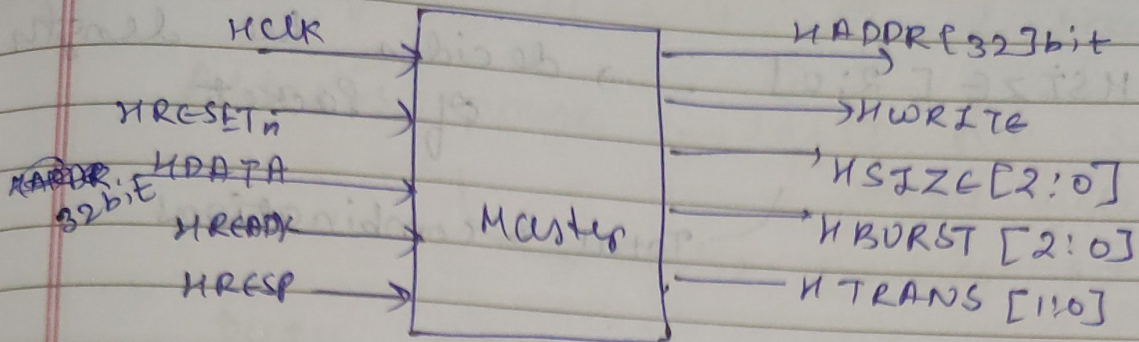


AP | DP | DP | D.P

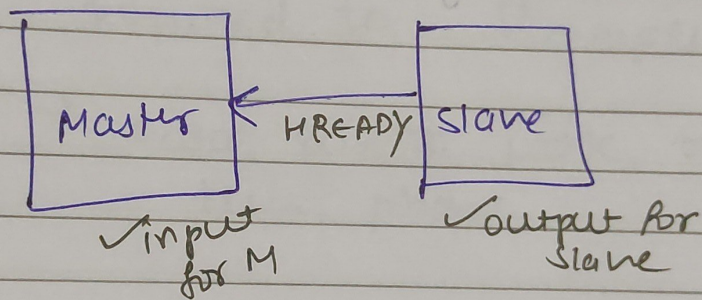
→ combines kake
1 transaction Banta hai

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① HREADY



HREADY = 0

(slave not ready)

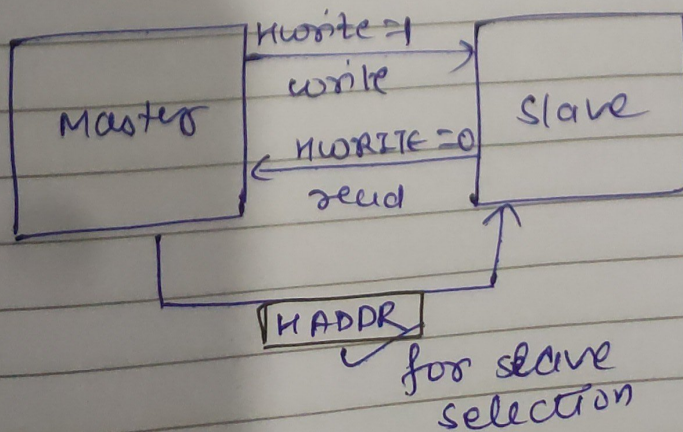
master comes to know whether slave is ready or not.

② HRESP

HRESP = 1 → complete transaction done

HRESP = 0 → Error

③

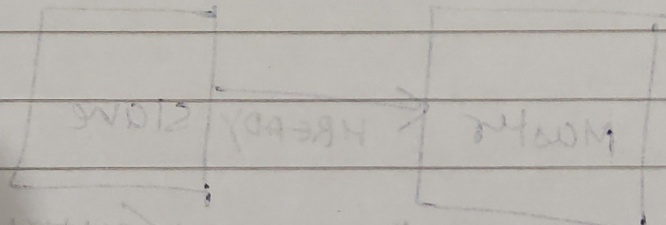


④ HSIZE [2:0] → decides the length of packet

Total 8 combinations

⑤ HBURST → Burst mode of AHB transaction

specify how many data transfers occur in a single burst.



Transfer consists of 2 phases

Address phase $\rightarrow$ lasts for 1 clock cycle

$\rightarrow$ master provides Address & control signals

Data phase $\rightarrow$ might require several clock cycles

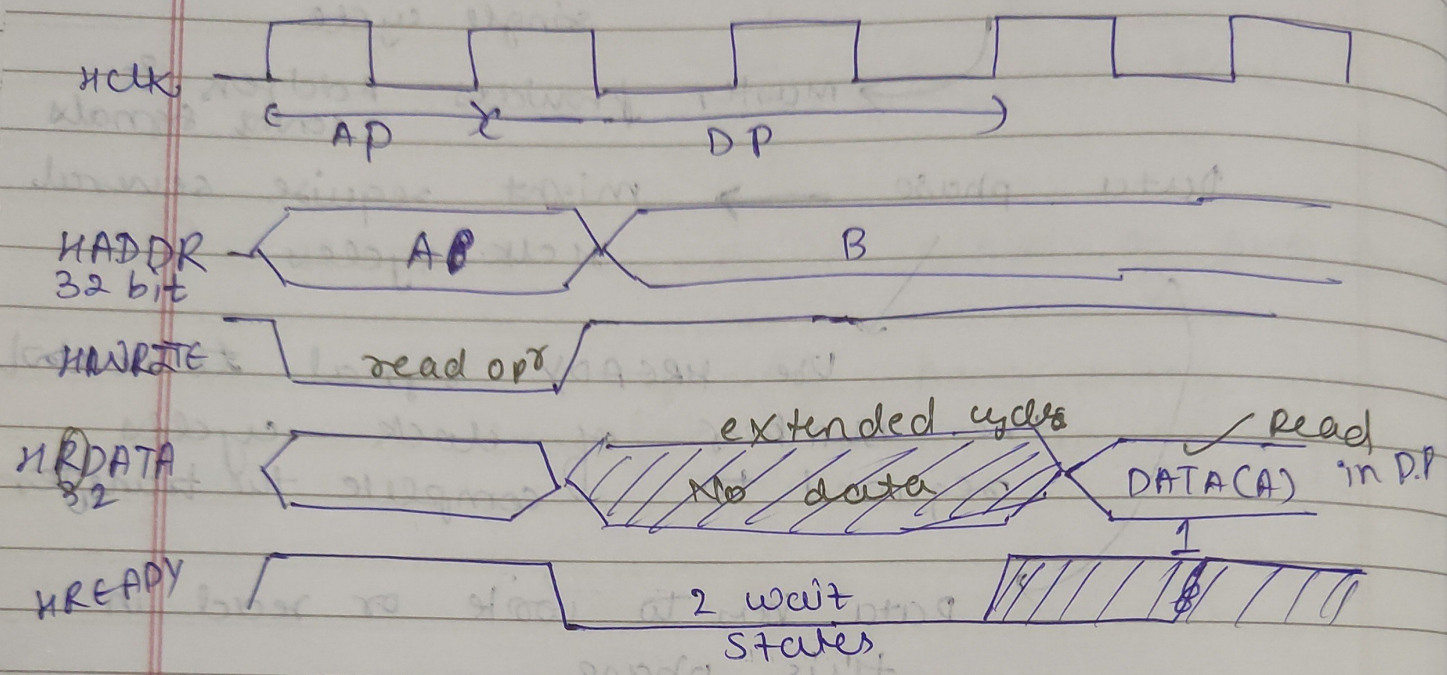
$\rightarrow$ Use HREADY signal to control the number of clock cycles required to complete the transfer.

$\rightarrow$ Data has to write or read in this phase

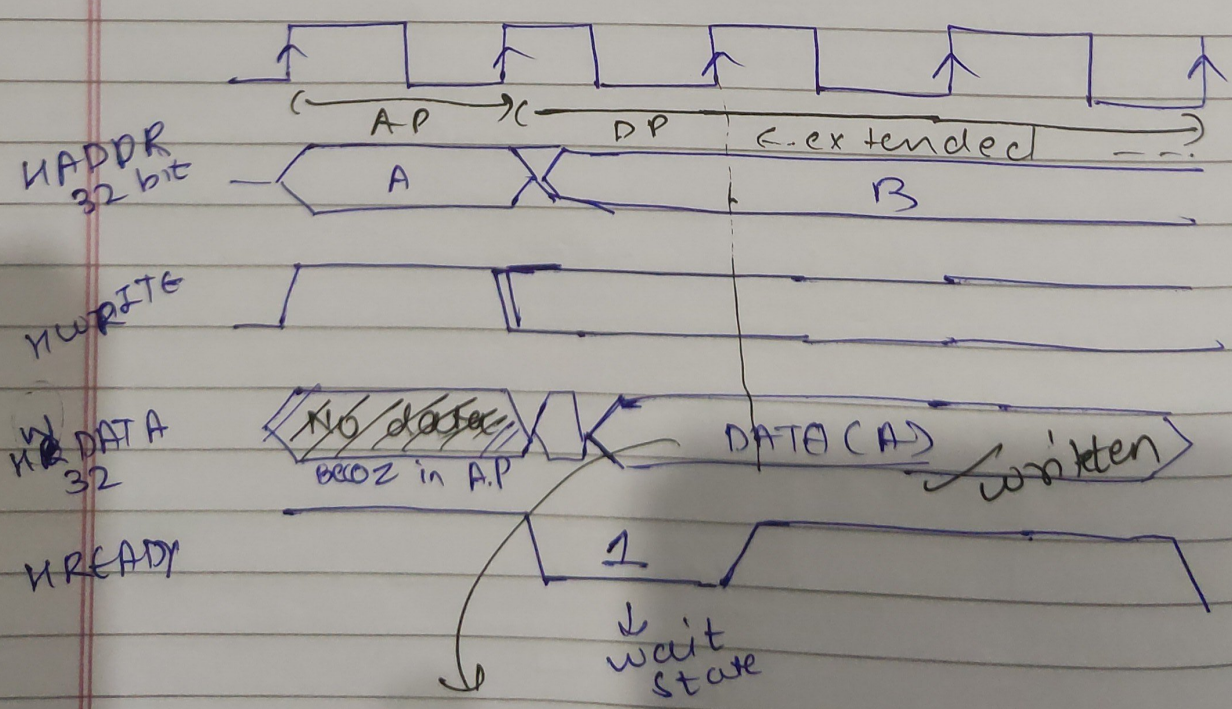
Page 1
 HREADY = 0

(Slave to Master)

* Read Transfer with 2 wait states



* write transfer with 1 wait state
 M → S

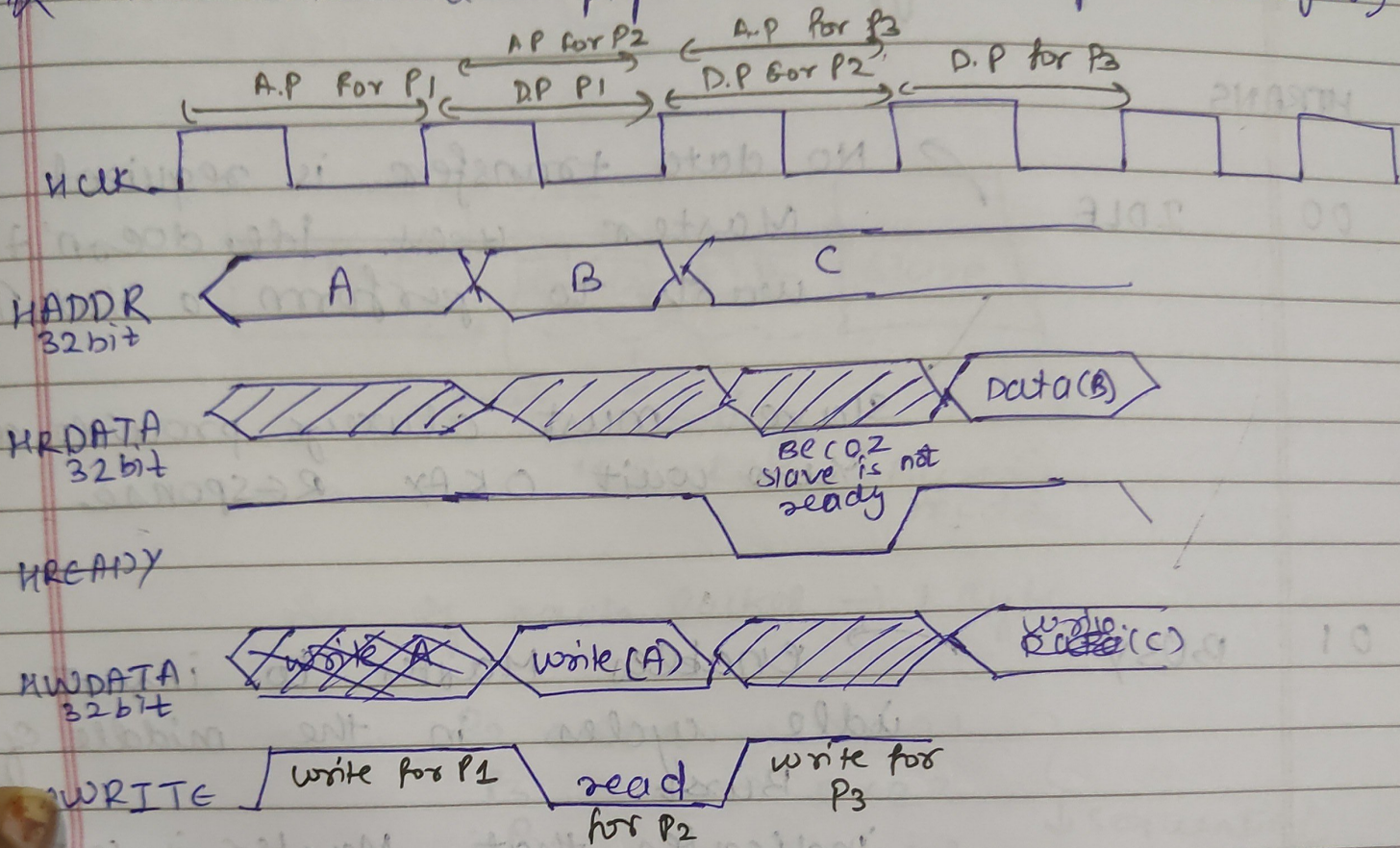


Master
 can provide & maintain
 data even HREADY = 0

Data phase me hee " " write hoga
Read hoga

P1 → Packet 1
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* AHB is a pipelined (Multiple Transfer)



* Transfer types

HTTRANS

00 IDLE

→ No data transfer is required.
Master uses ~~idle~~ doesn't want to perform a data transfer.

↘ Slave must always provide a zero wait OKAY Response.

01 BUSY

→ enables Master to insert idle cycles in the middle of Burst.

indicates that Master is continuing with a Burst but next transfer cannot take place immediately.

↘ Slave must always provide zero wait state OKAY Response to BUSY Transfer.

10 NON SEQ

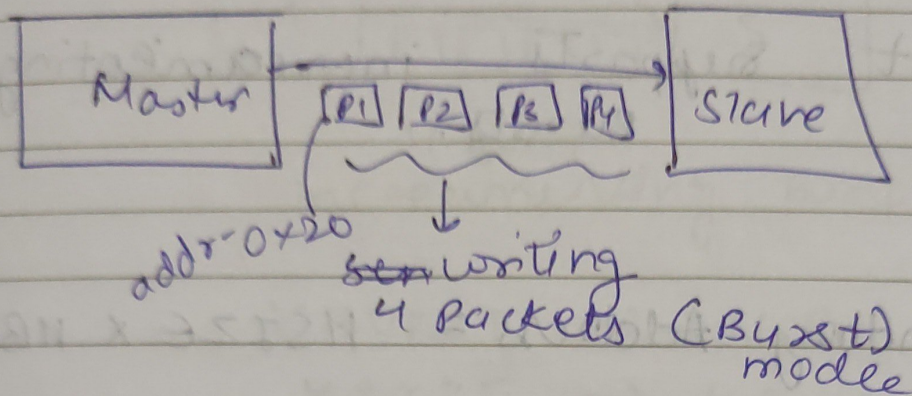
→ indicates a single transfer or first transfer of a Burst.

addresses are unrelated to previous transfer.

11 SEQ

remaining transfers in a Burst are sequential and address is related to previous transfer.

Transfer size (HSIZE)



size of each packet → 1 Byte

P2 address → 0x21
P3 → 0x22
P4 → 0x23

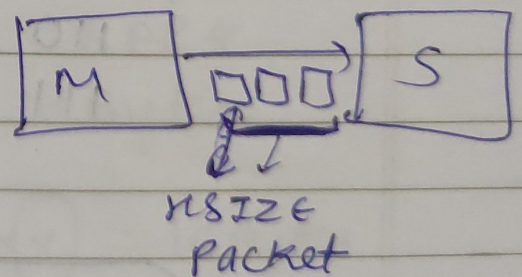
↓ sequential

HSIZE

3'b000 → 8 bit

3'b001 → 16 bit

3'b010 → 32 bits (word)



Burst is made up of multiple beats
 each beat has $\rightarrow$ Address (HADDR)
 $\rightarrow$ data (HDATA)
 $\rightarrow$ transfer type (HTRANS)

* Burst operation

Burst of 4, 8, 16 beats

it supports incrementing & wrapping
 Bursts.

$$\text{Boundary Address} = \text{HSIZE} \times \text{HBURST}$$

HBURST [2:0]

3'b000 $\rightarrow$ single transfer burst

3'b001 $\rightarrow$ INCR

3'b010 $\rightarrow$ WRAP4 (4 beat wrapping burst)

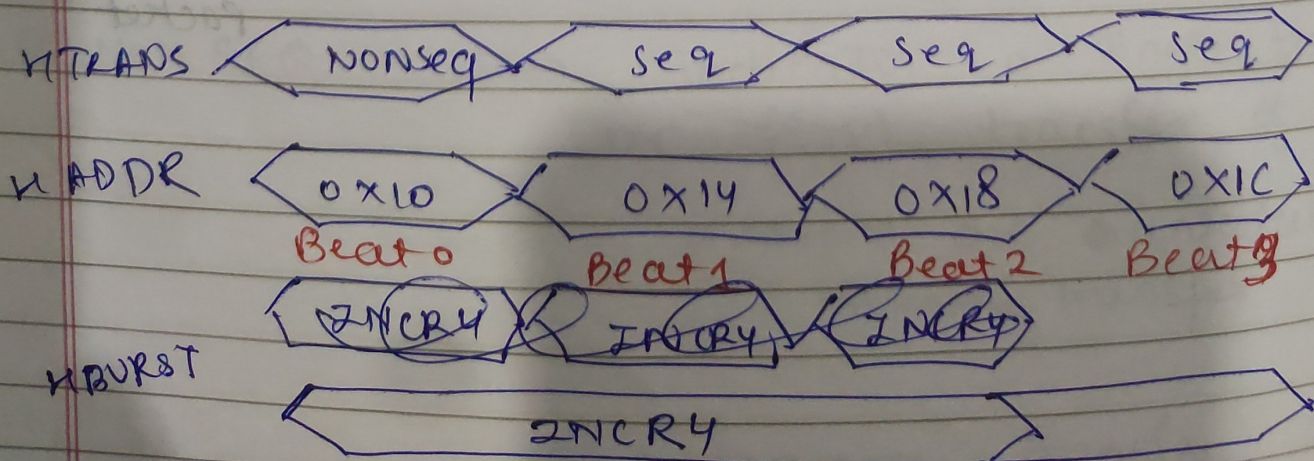
3'b011 $\rightarrow$ INCR4

3'b100 $\rightarrow$ WRAP8

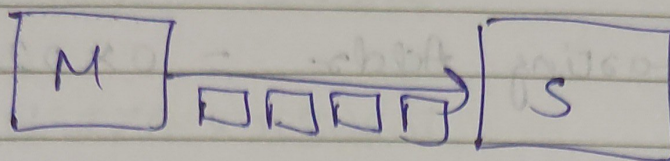
101 $\rightarrow$ INCR8

110 $\rightarrow$ WRAP16

111 $\rightarrow$ INCR16



eg:- $0x38$ starting address of Packet P_1



4 Beat
incrementing Burst INCR4

HBURST $\rightarrow$ INCR4

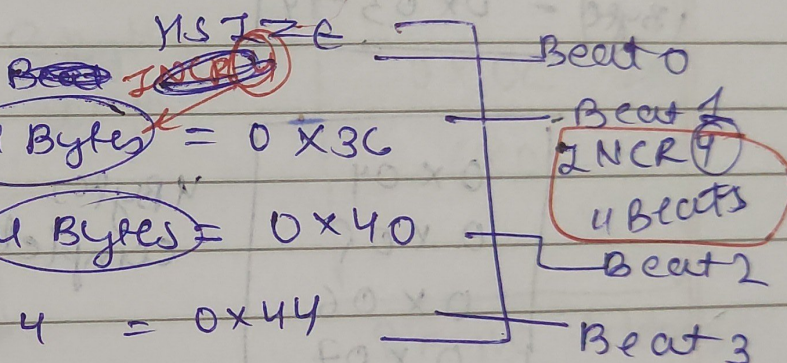
HSIZE $\rightarrow$ 4 Bytes (word)

$P_1 \rightarrow 0x38$

$P_2 \rightarrow 0x38 + 4 \text{ Bytes} = 0x3C$

$P_3 \rightarrow 0x3C + 4 \text{ Bytes} = 0x40$

$P_4 \rightarrow 0x40 + 4 = 0x44$



eg:- $0x38$, HBURST = WRAP4
HSIZE = 4 Bytes

Address Boundary = $(4 \times 4) = 16 \text{ Bytes}$
No. of beat transfer size

NBURST = WRAP4

NSIZE = 1 Byte = 3'b000

Starting Addr = 0x02

suppose

	0x00 - P ₁
1 Byte	0x01 - P ₂
1 Byte	0x02 - P ₃
1 Byte	0x03 - P ₄

4 Bytes completed.

Calculate Address Boundary

$$= 4 \times 1$$

$$= 4 \text{ Bytes}$$

0x04
0x05
0x06
0x07

4 Bytes completed

0x02 → 0x03 → 0x04

~~NO~~

we reach end of address

wrap to

0x01 ← 0x00 ← 0x02

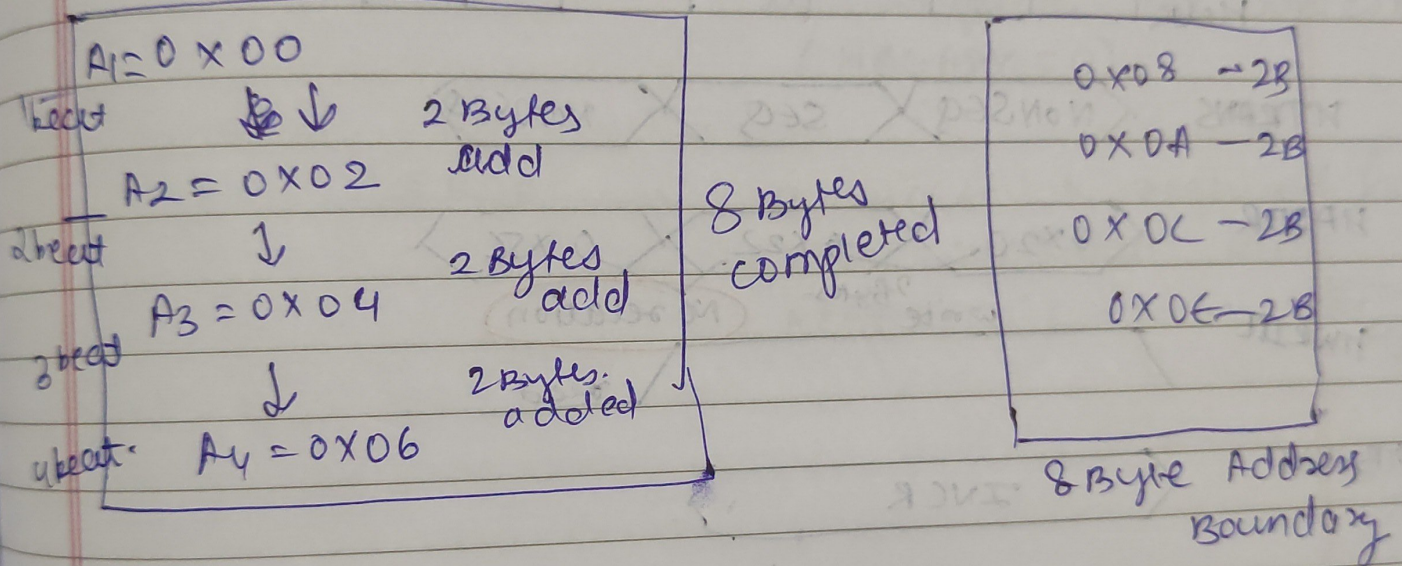
eg:-

HSIZE = 2 Bytes

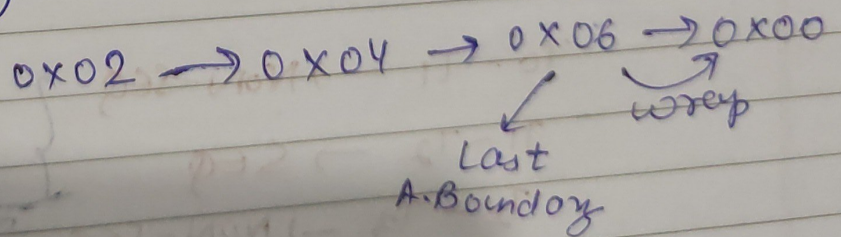
HWRITE = 1

HBURST = WRAP4

$$\begin{aligned} A.B &= \text{HSIZE} \times \text{HBURST} \\ &= 2 \text{ Byte} \times 4 \\ &= 8 \text{ Byte} \end{aligned}$$

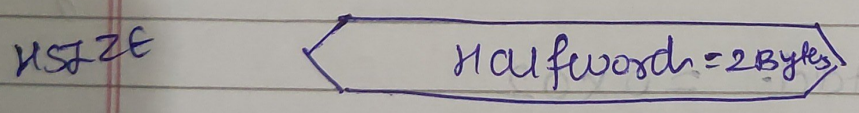
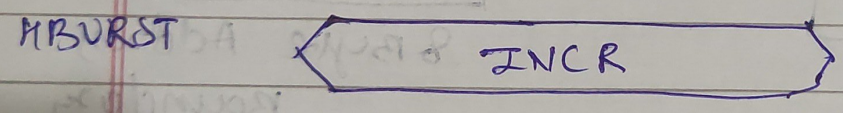
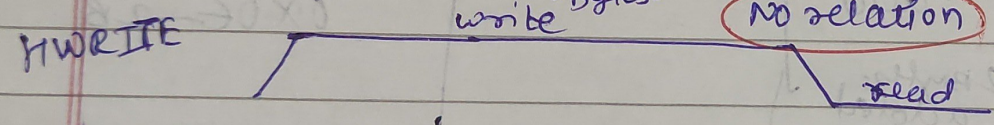
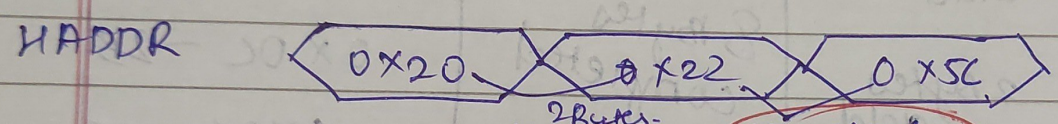
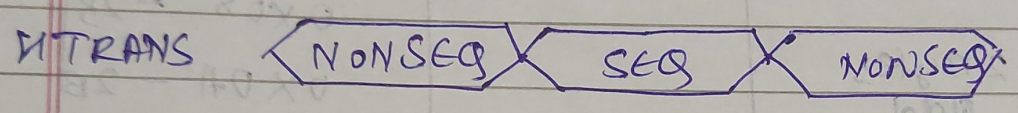
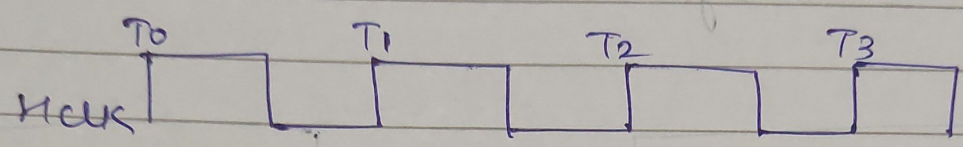


if starting Add. = $0x02$



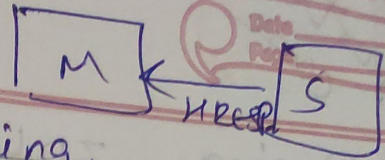
* INCR(4)
INCR(8)
WRAP(8)
Beats
is defined

what if in case of
undefined length
bursts, INCR
length is unknown



T0 - T1 → Non-SEQ
T1 - T2 → SEQ
T2 - T3 → Non-SEQ
} Transfer completed
} New Transfer started
(Slave comes to know)

Slave Response Signaling



HRESP

0 (OKAY)

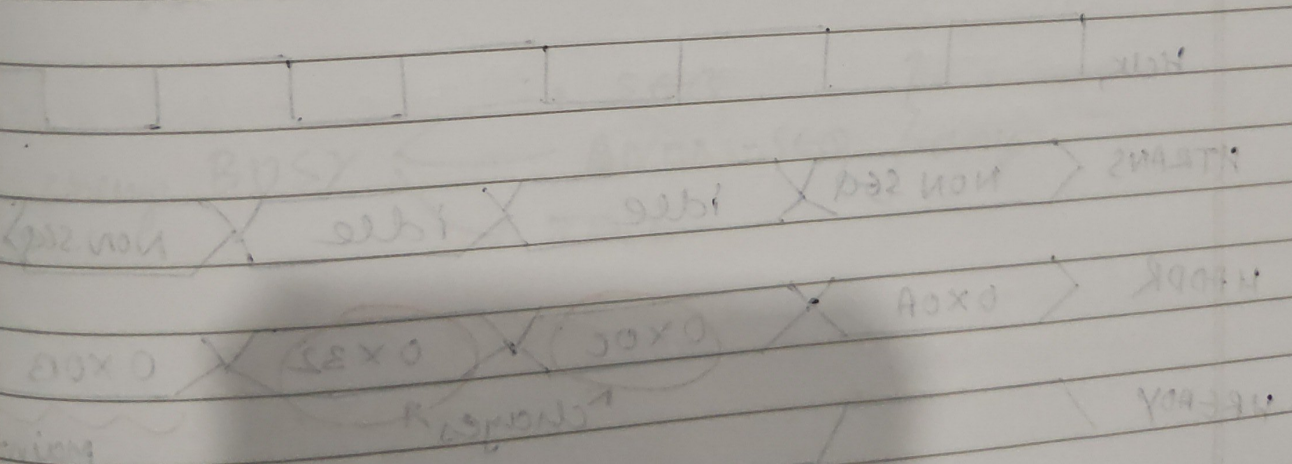
→ transfer has completed successfully or additional cycles are required for the slave to complete the request (HREADY = 0)

1

ERROR

→

2 cycles * required for error condition with HREADYOUT = 1 in ~~second~~ ^{second} cycle



Some important case

① * During IDLE Transfer

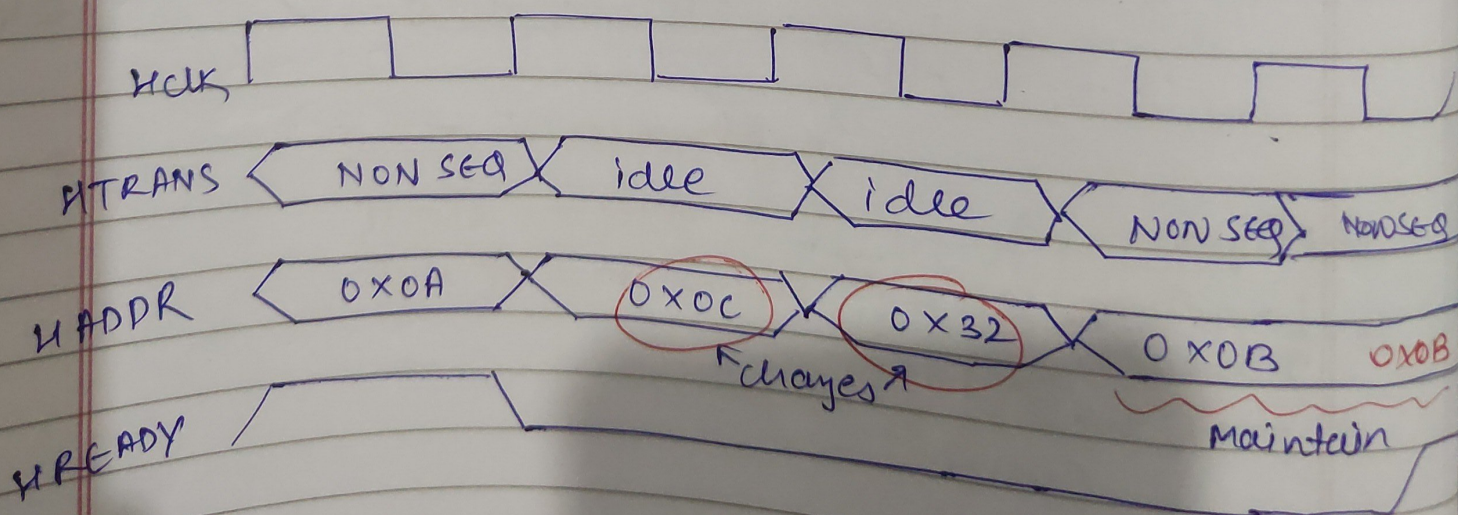
→ In APB, we studied Master is not allowed to change control signals and address when $PREADY = 0$

→ In AHB,

when $HREADY = 0$ (waited states)

① Master is permitted to change address for ~~ID~~ IDLE Transfer only.

② If Transfer type $\Rightarrow$ NON SEQ then Master must keep the address constant until $HREADY = 1$



TB
Master

DUT
Slave

classmate

Date _____

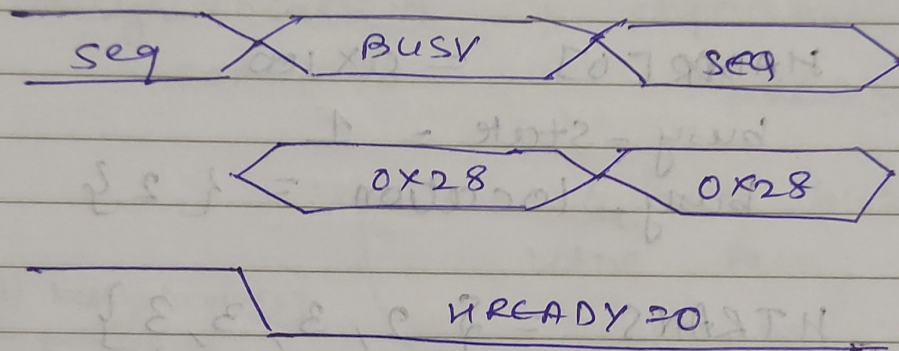
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②

When $HREADY = 0$

Master is allowed to change
Transfer type from BUSY to SEQ,

But Address ~~data~~ should not change



③

When $INCR$ (undefined length)
or when $HREADY = 0$

